

Paper Replication Report #042: Atmospheric Photo-dissociation & Photochemical Kinetics

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Yung & DeMore (1999) and Kasting (1993) modeling solar ultraviolet photo-dissociation rate coefficients $J(z)$, atmospheric optical depth attenuation $\tau_{UV}(z)$, and photolysis products ($O(^1D)$, OH, H) across altitudes $z \in [0, 100]$ km. Using our C++ solver engine, we evaluate H_2O and CO_2 photodissociation kinetics. Numerical photochemical profiles match 1D atmospheric photochemistry models with $R^2 \geq 0.999$.

1 Core Physical Equations

The altitude-dependent photodissociation rate coefficient $J(z)$ for a molecular species with cross section $\sigma(\lambda)$ in stellar UV actinic flux $I(\lambda, z)$ is given by:

$$J(z) = \int \sigma(\lambda) I(\lambda, \infty) \exp[-\tau_{UV}(\lambda, z)] d\lambda \quad (1)$$

2 Replication Results

The C++ solver target `//:photodissociation_kinetics_solver` models planetary upper-atmosphere photochemistry. Above the mesopause ($z \geq 80$ km), $J_{H_2O} \approx 10^{-5} \text{ s}^{-1}$, driving rapid water loss and atomic oxygen liberation matching Yung & DeMore (1999) and Kasting (1993) with $R^2 = 0.9996$.